

Hotel Reservation Management System

1. System Overview

The Hotel Reservation Management System allows users to book hotel rooms by selecting check-in date, check-out date, number of guests, and preferred services such as rooms and spa. The system validates availability, calculates cost, and confirms the reservation.

2. Pseudocode

```
START
Display Home Screen
User selects Check-in Date
User selects Check-out Date
User enters Number of Guests

IF dates are valid THEN
    Display available rooms
    User selects room type
    User selects additional services (optional)

    Calculate total cost
    Display booking summary

    IF user confirms booking THEN
        Save reservation details
        Display booking confirmation
    ELSE
        Cancel booking
    ENDIF
ELSE
    Display error message
ENDIF
END
```

3. Algorithm / Flowchart Description

- Start the system
- Display the hotel booking interface
- Input check-in, check-out, and guest count
- Validate dates and guest capacity
- Check room availability
- Allow user to choose room and services
- Compute total price
- Confirm booking
- Store reservation and notify user

- End process

4. Program Logic Explanation

The program starts by showing the main booking interface. The user inputs the check-in and check-out dates along with the number of guests. The system checks if the dates are valid and ensures room availability. Once validated, the user selects a room type and optional services like spa. The system calculates the total cost based on the selections. If the user confirms the booking, the reservation is saved and a confirmation message is displayed. If not, the booking process is cancelled.